

useState

```
const [state, setState] = useState(initialValue);
```

state -> current value

setState -> current value update

useReducer -

```
const [state, dispatch] = useReducer(reducerFn, initialState);
```

state -> current value

reducer (function) -> A pure function that takes the current state and an action, then returns a new state.

(state, action) => new state

action -> An object that describes what happened. It is the only way to trigger a state change. It usually contains a type (must) and optionally a payload (data).

type -> A String inside the action object.

payload -> Extra data needed to update the state.

dispatch -> The function you call to "send" an action to the reducer.

```
const initialState = {
```

```
  count: 0
```

```
  username: ""
```

```
}
```

```
const reducerFn = (state, action) => {
```

```
  switch(action.type) {
```

```
    case "INCREMENT":
```

```
      return {...state, count: count+1}
```

```
    case "DECREMENT":
```

```
      return {...state, count: count-1}
```

```
    case "USERNAME":
```

```
      return {...state, username: action.payload}
```

```
    default:
```

```
      return state
```

```
  }
```

```
}
```

```
const [state, dispatch] = useReducer(reducerFn, initialState);
```

```
log(state); // {count: 1, username: ""}
```

```
<button onClick={dispatch({type: "INCREMENT"})}>increment</button>
```

```
log(state); // {count: 1, username: ""}
```

```
<button onClick={dispatch({type: "USERNAME", payload: name})}>increment</button>
```

```
log(state); // {count: 1, username: "name"}
```

